

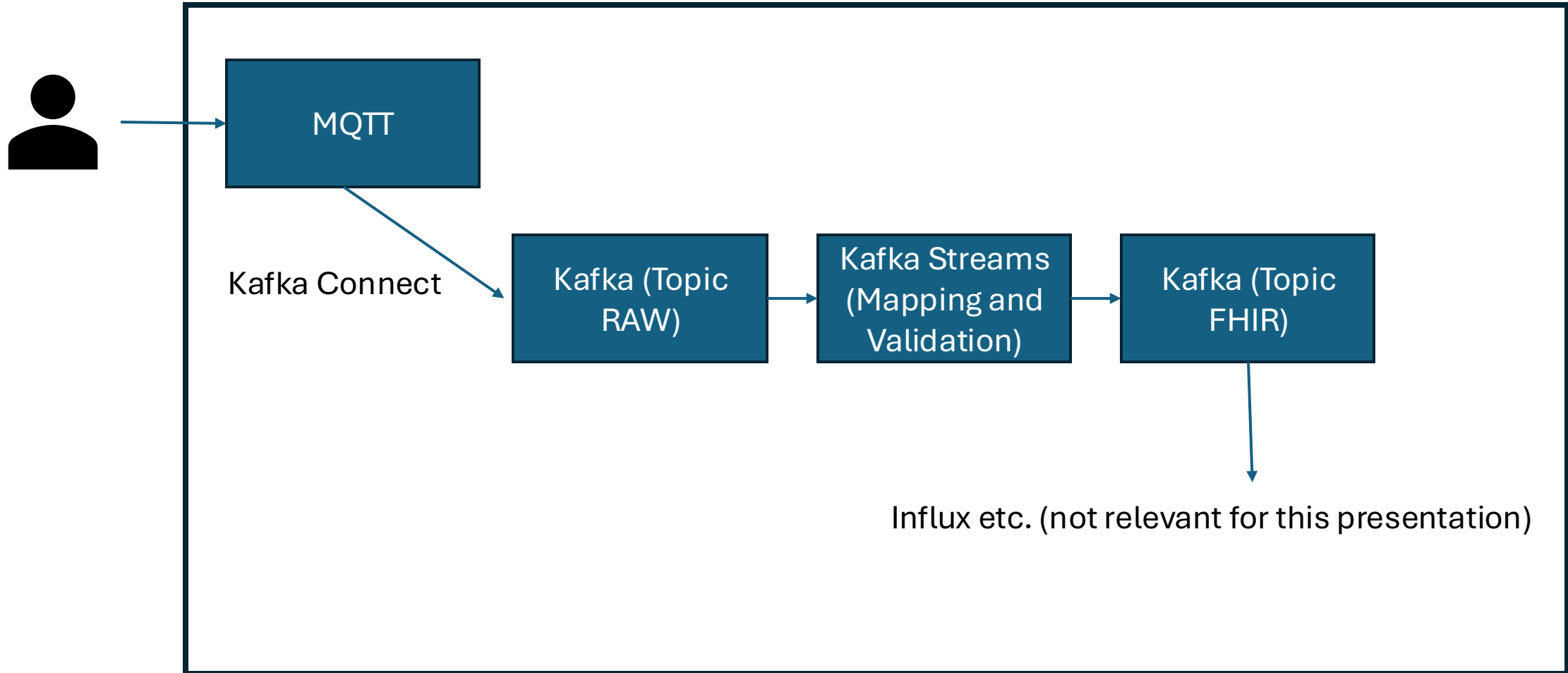
How to handle FHIR mapping

Async presentation of my research

Assumptions

- I will summarize the suggested new approaches on the slides
- I will make the changes one-by-one on the current prototype implementation, but of course they can be combined differently
- I researched the approaches like I understood them, please let me know if I misunderstood your ideas

Current approach

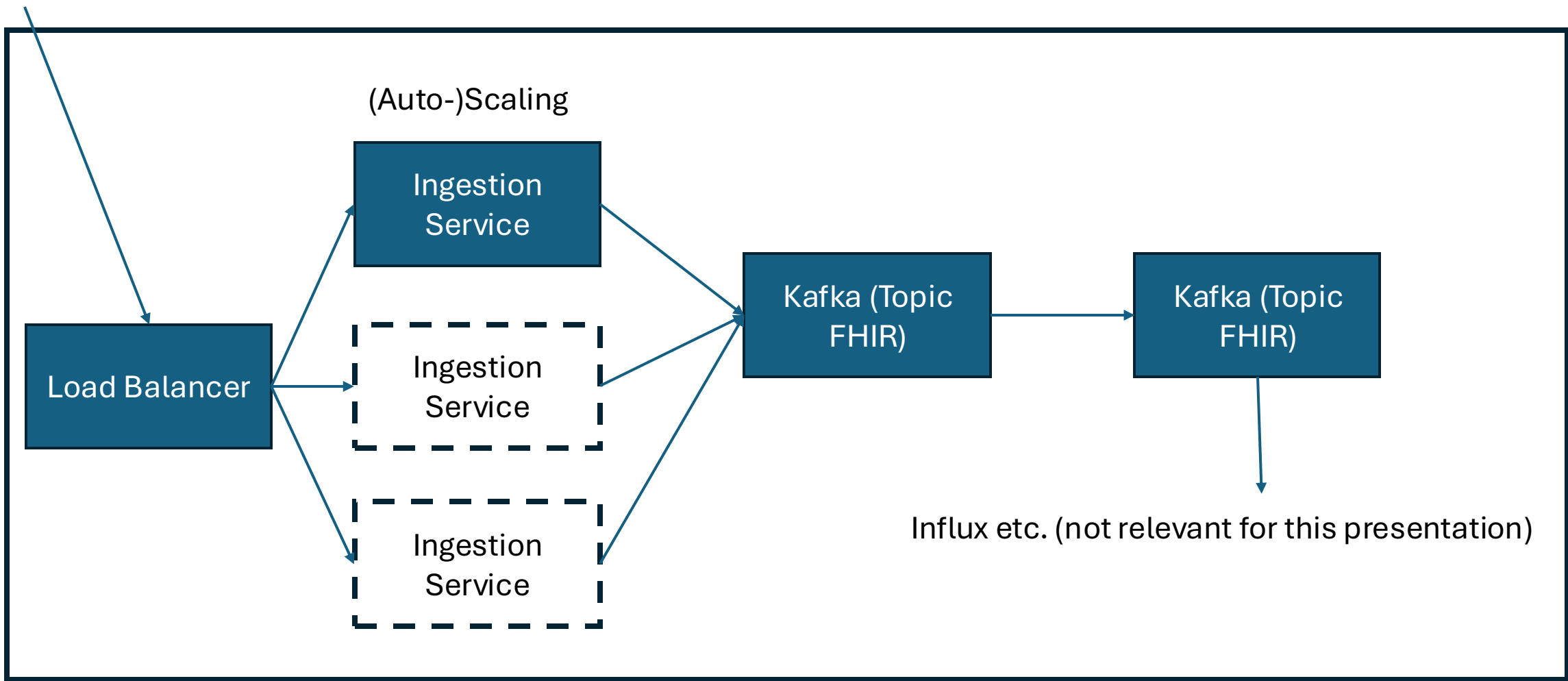


Current approach

- For the current implementation please [check out the repo](#)



New: Ingestion Service



New: Ingestion Service

- Pro

1. Reduced number of services
2. Authentication made easy (Oauth, JWT)

- Con

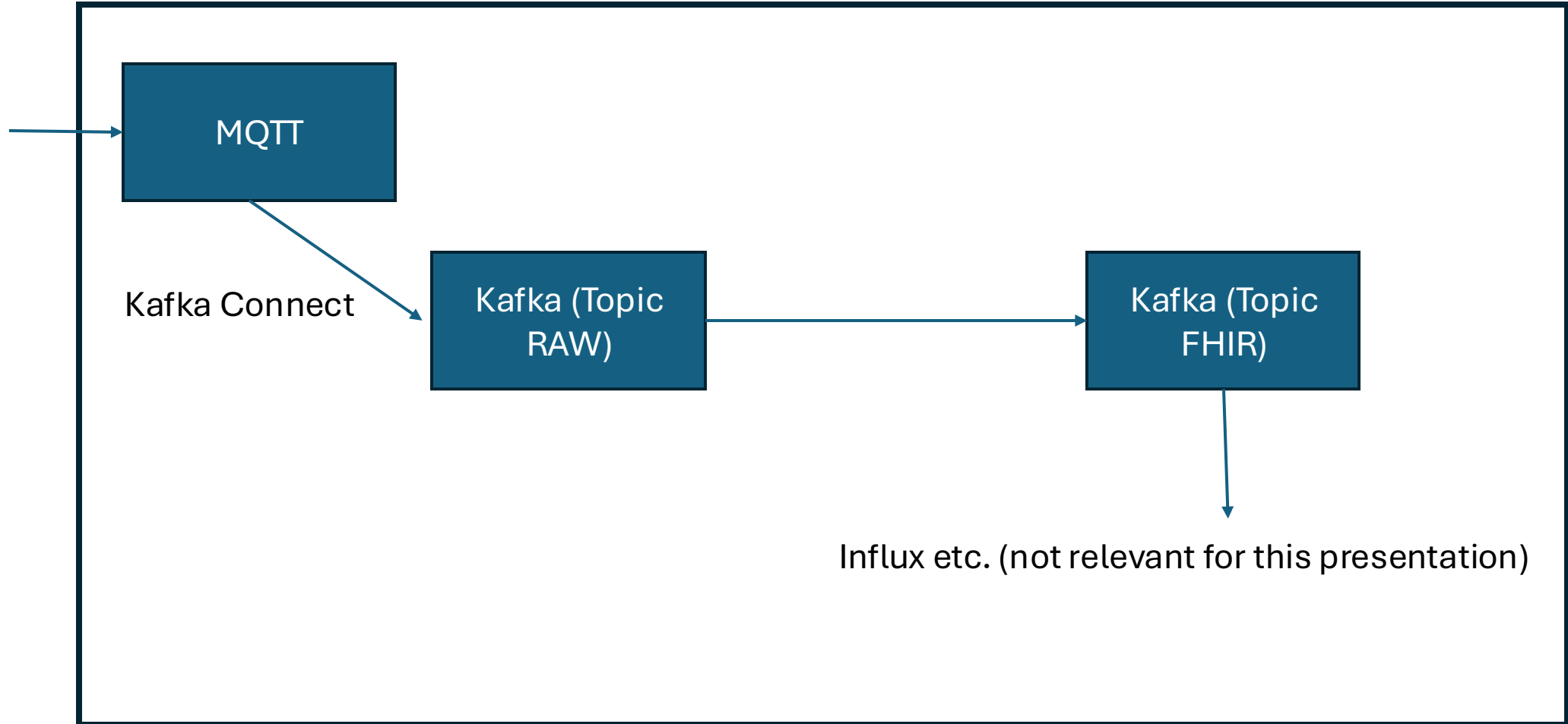
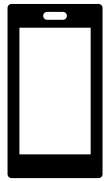
1. We need to consider a load balancer (how is load distributed intelligently)
2. We need to consider scaling logic (via k8s?)
3. We need to consider failure handling (retries, backoff, at-least-once delivery)

New: Ingestion Service

- Personal thoughts
 - When Charité implements this, they will probably want to implement their own authentication anyways
 - The MQTT broker in the original approach would already be behind a firewall

New: On-device mapping via libraries

Mapping
happens
here



New: On-device mapping via libraries

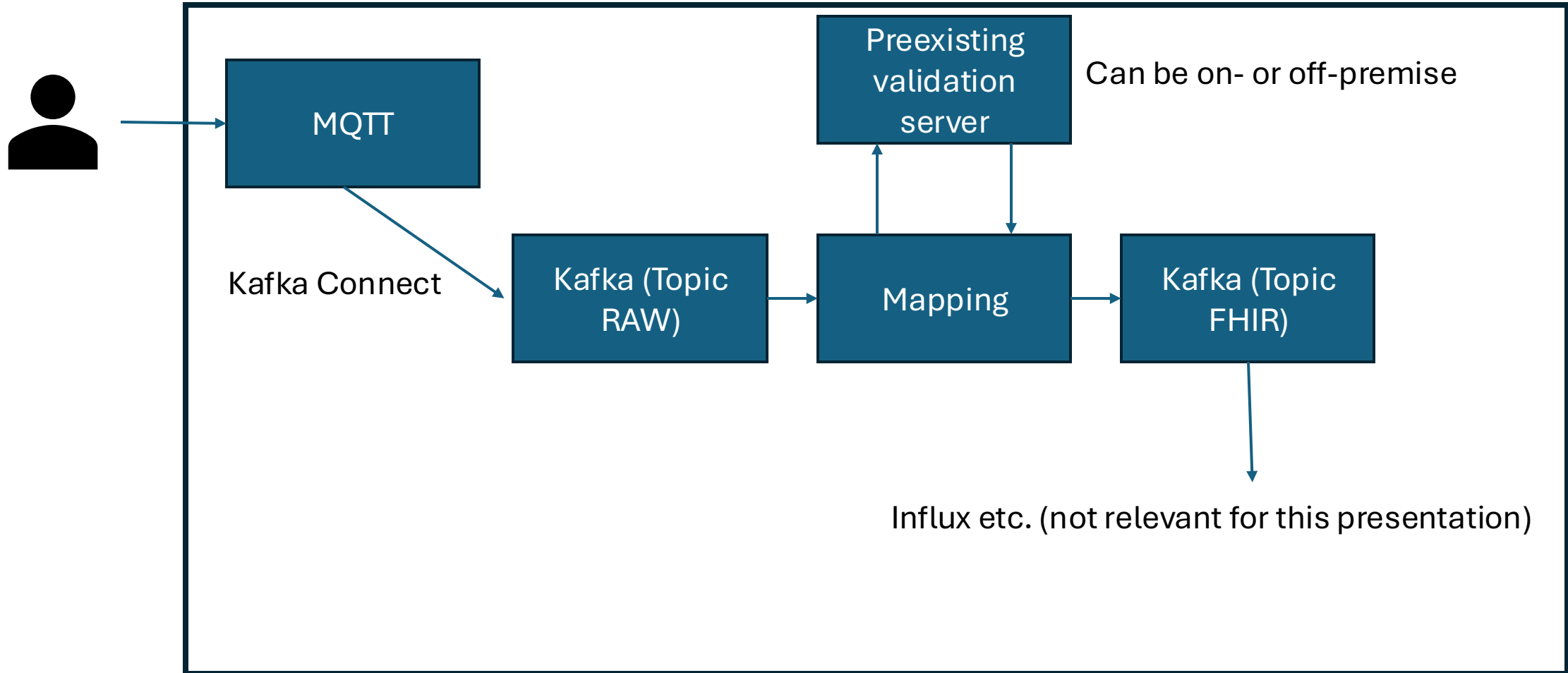
- Pro

1. Reduces complexity in k8s-cluster
2. On-prem system only ever encounters valid FHIR
3. Preexisting libraries to handle mapping and validation

- Con

1. We lose control, since mapping doesn't happen on-prem
2. Debugging gets harder in production since we need to push app-updates
3. Consider the target demographic for continuous monitoring and their digital literacy
4. Failure handling has to be done on the device in limited computation time

New: FHIR server for validation



New: FHIR server for validation

- I don't really see the advantages of this approach, **since validation is not our main concern** / pain-point but rather mapping
- Con:
 1. We don't have control over the exact validation used (Libraries that are validated against, behavior in case of failure etc.)
 2. External dependencies for the mapping service (no matter if on-prem or not) -> doesn't really fit with the pattern of statelessness

My Suggestion

- Honestly: Staying with the current approach
- Authentication is not our concern I think since Charité probably has their own systems and are not interested in increasing complexity by integrating a new service
- MQTT does not need to be scaled for our load (less complexity)
- On-device mapping makes production changes more work-intensive

My Suggestion

- Letting the client be as thin as possible gives us the most direct control over the system
- FHIR servers for validation increase number of network hops and in the end they do exactly what the current validator does